Ekaterina ILIN

astrophysicist specializing in star-planet interactions

Academic positions

2026 →ongoing — **Max-Planck Research Group Leader @ Max-Planck Institute for Solar System Research**
Magnetic Interaction of (Sub)Stellar Companion Systems

2023 →2026 — **Postdoc @ Netherlands Institute for Radio Astronomy (ASTRON), Netherlands**
Project STORMCHASER; PI: H. K. Vedantham

2022 →2023 — **Postdoc @ Leibniz Institute for Astrophysics Potsdam (AIP), Germany**
Funded by the German Aerospace Center (Deutsches Zentrum für Luft und Raumfahrt); PI: **E. Ilin**

Education

2018 →2022 — **PhD** in Astrophysics **@ Leibniz Institute for Astrophysics Potsdam (AIP), Germany**
Supervisor: Prof. Dr. Katja Poppenhäger
High lights: stellar flares as probes of magnetism in stars and star-planet systems
"summa cum laude"

→ I specialize in the magnetism of small stars, its origins, long-term evolution, and interaction with the planets in their orbit.

2016 →2018 — **Master** of Science in Astrophysics **@ University of Potsdam, Germany**
Supervisor: Prof. Dr. Klaus Strassmeier, mentor: Dr. Sarah J. Schmidt
Flares in clusters using K2 data
1,0 "with distinction"

2012 →2016 — **Bachelor** of Science in Physics **@ Karlsruhe Institute for Technology, Germany**
Specialization in biophysics, Supervisor: Prof. Dr. Ulrich Nienhaus
Uptake of TAT-coated gold nanoclusters in live HeLa cells
1,4 "very good"

First author refereed publications

2 key publications highlighted

Searching for planet-induced radio signal from the young close-in planet host star HIP 67522
Ilin, E.; Bloot, S.; Callingham, J. R.; Vedantham, H. K.
2025, Astronomy & Astrophysics, 699, A147 doi:10.1051/0004-6361/202554684 arXiv:2507.00796

Close-in planet induces flares on its host star
Ilin, E.; Vedantham, H. K.; Poppenhäger, K.; Bloot, S.; Callingham, J. R.; Brandeker, A.; Chakraborty, H.
2025, Nature, 643, 645-648 doi:10.1038/s41586-025-09236-z arXiv:2507.00791
Detected the first confident case of planet-induced flaring from time-domain observations with TESS and CHEOPS through Bayesian analysis of flare clustering near the innermost planet's transit.

The corona of a fully convective star with a near-polar flare
Ilin, E.; Poppenhäger, K.; Stelzer, B.; Dsouza D.
2024, Astronomy & Astrophysics, 687/A138 doi:10.1051/0004-6361/202449541 arXiv:2405.05580

Planetary Perturbers: Flaring star-planet interactions in Kepler and TESS
Ilin, E.; Poppenhäger, K.; Chebly, J.; Ilić N.; Alvarado-Gómez, Julián D.
2023, Monthly Notices of the Royal Astronomical Society, 523/3, pp 4326–4339 doi:10.1093/mnras/stad3398 arXiv:2311.04316

Flaring Latitudes in Ensembles of Low Mass Stars
Ilin, E.; Angus, R.; Luger, R.; Morris, B.; Jehn, F.U.
2023, Monthly Notices of the Royal Astronomical Society, 523/3, pp 4326–4339 doi:10.1093/mnras/stad1690 arXiv:2306.02007

Searching for star-planet interactions in AU Mic TESS observations
Ilin, E.; Poppenhäger, K.
2022, Monthly Notices of the Royal Astronomical Society, 513/3, pp 4579-4586 doi:10.1093/mnras/stac1232 arXiv:2204.14090

Localizing flares to understand stellar magnetic fields and space weather in exo-systems
Ilin, E.; Poppenhäger, K.; Alvarado-Gómez, J. D
2021, Astronomische Nachrichten, doi:10.1002/asna.20210111 arXiv:2112.09676

Giant white-light flares on fully convective stars occur at high latitudes
Ilin, E.; Poppenhäger, K.; Schmidt, S. J.; Järvinen, S. P.; Newton, E. R.; Alvarado-Gómez, J. D; Pineda, S. J.; Davenport, J. R. A.; Oshagh, M.; Ilyin, I.
2021, Monthly Notices of the Royal Astronomical Society, 507/2, pp 1723–1745 doi:10.1093/mnras/stab2159 arXiv:2108.01917
First determination of near-polar flaring in four rapidly rotating M dwarfs. Developed new method for localizing flare footpoints on low mass stars using optical time-domain observations and high-resolution optical spectroscopy.

AltaiPony – Flare science in Kepler, K2 and TESS light curves
Ilin, E.
2021, Journal of Open Source Software, 6(62), 2845 10.21105/joss.02845
Peer-reviewed open source python toolbox for statistical flare analysis in time-domain photometry. Used in > 30 refereed publications (2025).

Flares in Open Clusters with K2. II. Pleiades, Hyades, Praesepe, Ruprecht 147, and M67
Ilin, E.; Schmidt, S. J.; Poppenhäger, K.; Davenport, J. R. A.; Kristiansen, M. H.; Omohundro, M.
2021, Astronomy & Astrophysics, 645, A42, 25 pp. doi:10.1051/0004-6361/202039198 arXiv:2010.05576

Flares in Open Clusters with K2. I. M45 (Pleiades), M44 (Praesepe) and M67
Ilin, E.; Schmidt, S. J.; Davenport, J. R. A.; Strassmeier, K. G.
2019, Astronomy & Astrophysics, 622, A133/16. doi:10.1051/0004-6361/201834400 arXiv:1812.06725

Refereed publications with significant contribution, or as supervisor

Searching for gamma-ray emission from stellar flares
Song Yuzhe; Paglione Timothy A. D., **Ilin E.**
2024, Monthly Notices of the Royal Astronomical Society, 531/3, 3215-3221 doi:10.1093/mnras/stae1347 arXiv:2405.15435

Lower than expected flare temperatures for TRAPPIST-1
Maas A. J.; **Ilin E.**; et al.
2022, Astronomy & Astrophysics, 928/2, 147. doi:10.1051/0004-6361/20224386 arXiv:2210.11103

Refereed publications with minor contributions

Occurrence rate of stellar Type II radio bursts from a 100 star-year search for coronal mass ejections
Konijn, D.; ...; **Ilin, E.**; ...; et al.
2025, Astronomy & Astrophysics, 703, id.A198, 13 pp. doi:10.1051/0004-6361/202554317 arXiv:2511.09296

Polarized, variable radio emission from the scallop-shell binary system DG CVn
Kaur, S.; ...; **Ilin, E.**; ...; et al.
2025, Astronomy & Astrophysics, 701, A69 doi:10.1051/0004-6361/202555222 arXiv:2507.09366

Hints of auroral and magnetospheric polarized radio emission from the scallop-shell star 2MASS J05082729-2101444
Kaur, S.; ...; **Ilin, E.**; ...; et al.
2024, Astronomy & Astrophysics Letters, 691/L17 9 pp. doi:10.1051/0004-6361/202452037 arXiv:2410.22449

Seaweed as a resilient food solution in nuclear winter
Jehn F. U.; Jasmine Dingal, F.; Mill, A.; **Ilin, E.**; Harrison, C.; Roleda, M. Y. ; Denkenberger, D.
2024, Earth's Future, 12/1 doi:10.1029/2023EF003710

Three young planets around the K-dwarf K2-198: high-energy environment, evaporation history, and expected future
Ketzer, L.; Poppenhaeger, K.; Baratella, M.; **Ilin, E.**
2024, Monthly Notices of the Royal Astronomical Society, 527/1, pp. 374-385 doi:10.1093/mnras/stad3197 arXiv:2311.06897

Focus of the IPCC Assessment Reports has shifted to lower temperatures
Jehn F. U.; Kemp L.; **Ilin E.**; Funk C.; Wang J. R.; Breuer L.
2022, Earth's Future, 5/10 doi:10.1029/2022EF002876

Simulating the Space Weather in the AU Mic System: Stellar Winds and Extreme Coronal Mass Ejections
Alvarado-Gómez, J. D; Cohen O.; Drake J. J.; Fraschetti F.; Poppenhäger, K.; Garraffo C.; Chebly J.; **Ilin, E.**; Harbach

L.; Kochukhov O.
2022, The Astrophysical Journal, 928/2, 147. doi:10.3847/1538-4357/ac54b8 arXiv:2202.07949

Teaching and supervision

total supervised: 3 Master level, 2 Bachelor level

2025 →2026 **Supervising students @ Kapteyn Institute** Groningen, NL
Supervised the BSc project of Andrei Danila, and the MSc project of Casper Hindriks.

2024 **Supervising summer student @ ASTRON** Dwingeloo, NL
Supervising a 10-week summer student project with undergraduate student Necdet Canim (Pomona College, CA, USA).

2024 **Invited lecturer at summer school @ ICE-CSIC Institute of Space Sciences** Barcelona, Spain
On the role of flares in star-planet interactions

2023 **Lecturer at summer school @ German National Scholarship Foundation** Vienna, Austria
Teaching undergraduates about exoplanets in an interdisciplinary context

2023 **Invited lecturer @ Tag der Schulastronomie Sachsen** Meißen, Germany
Lecture about exoplanets for high school physics teachers in Saxonia.

2022 →2025 **Mentoring graduate student Cinta Vidante @ University of Potsdam ()** Potsdam, Germany
Graduated with distinction; publication in prep.

2020 →2022 **Mentoring graduate student Aaron Maas @ University of Heidelberg** remote
Maas, Ilin et al. (2022). Now PhD student in astrobiology at CSIC-ICE.

2020 **Teaching assistant @ University of Potsdam** Potsdam, Germany
Galaxies and Cosmology undergraduate course

2016 →2018 **Teaching assistant @ University of Potsdam** Potsdam, Germany
Introduction to mathematical methods in physics

External funding

total ca. 4,700,000€

2027 →ongoing **PI of ERC Starting Grant** PLANETARE
Project funding, 1,500,000€

2026 →ongoing **PI of Max-Planck Research Group** Max-Planck Society
Junior research group funding, 2,750,000€

2022 →2023 **PI of DLR Grant "XrayLoops"** Deutsches Zentrum für Luft- und Raumfahrt (DLR)
Two years of postdoctoral funding, ca. 140,000€

2021 →2022 **Fulbright Scholarship @ American Museum of Natural History** Fulbright Commission
Awarded to doctoral candidates. Includes a stipend, travel grant, and academic support throughout a six month research stay in the USA, ca. 12,000€

2020 →2022 **Doctoral Scholarship** German Academic Scholarship Foundation
Includes a stipend, travel grants, and academic support throughout the doctoral studies, ca. 55,000€

2012 →2018 **Full Scholarship** German Academic Scholarship Foundation
Includes a stipend, travel grants, and academic support throughout the undergraduate and graduate studies, ca. 60,000€

Awards and prizes

2024 **Stargazer Award 2024**
5k€ awarded to outstanding PhD theses in astronomy in German speaking countries.

2024 **Publication Prize @ Leibniz-Kolleg Potsdam**
2.5k€ awarded to outstanding PhD theses in the natural science in Berlin and Brandenburg.

2019 **Graduation Award @ Physikalische Gesellschaft zu Berlin, Berlin, Germany**
1.5k€ awarded to excellent master theses in physics earned at universities in Berlin and Potsdam.

2014 **Erasmus+ Scholarship @ Universitet i Oslo, Norway**
Awarded funding for six months of undergraduate studies. Focus: space physics, computational physics

Observing grants as Principal Investigator

2026 **La Silla 3.6m telescope / HARPSpol, 28h**
PI: **E. Ilin**; Co-Is: Stefano Bellotti, Aline Vidotto, Alexis Lavail, Pascal Petit

2025 **XMM-Newton Large Program, 612ks Priority A**
PI: **E. Ilin**; Co-PIs: K. Poppenhäger, R. Osten, A. Strugarek; Co-Is: Nikoleta Ilić, H. K. Vedantham

2025 **Submillimeter Array, 10 hours DDT**
PI: **E. Ilin**, Co-I: J. Lovell

2025 **CHEOPS, 270 orbits, Priority 1**
PI: **E. Ilin**

2024 **VLT/UVES, 6.4 hours**
PI: **E. Ilin**, Co-Is: H. K. Vedantham, S. Järvinen

2024 **Australian Telescope Compact Array, 144 hours**
PI: **E. Ilin**, Co-Is: S. Bloot, J. Callingham

2023 **CHEOPS, 128 orbits, Priority 1**
PI: **E. Ilin**, Co-Is: K. Poppenhäger

2021 **XMM-Newton, 36ks priority A**
PI: **E. Ilin**, Co-Is: K. Poppenhäger, B. Stelzer

Observing grants as Co-Investigator

2025 **GMRT, 32 hours**
PI: S. Kaur, Co-Is: D. Vigano, **E. Ilin** and 14 others

2020 **SALT HRS, 2.8 hours DDT**
PI: K. Poppenhäger, Co-Is: **E. Ilin**, S. J. Schmidt

Invited talks at conferences

2026 **Radio2026** Bielefeld, Germany
Radio Cool Stars

2026 **COSPAR** Florence, Italy
Magnetic star-planet interactions: Teenage planets under the influence of their host stars

2025 **EAS Annual Meeting** Cork, Ireland
Review: Magnetic star-planet interactions

2024 **NOVA Symposium** Den Haag, NL
Magnetic winds and plasma seas: Interactions in star-planet systems

2023 **All-hands-on-deck meeting – German Exoplanet Diversity Group** Munich, Germany
How close-in planets manipulate their host stars

2023 **From the Heliosphere to Astrospheres - Lessons for Exoplanets and their Habitability** Bad Honnef, Germany
Magnetic star-planet interactions

Invited colloquia and seminar talks (selection)

2025 **St. Mary's University** Halifax, CA
Magnetic star-planet interactions

2025 **University of Geneva** Geneva, CH
Magnetic star-planet interactions: Teenage planets under the influence of their host stars

2025 **Leiden Observatory** Leiden, NL
Self-roast: Close-in planet induces flares on its host star

2025 **Vienna University** Vienna, AT
Channeling sparks through interplanetary space: Magnetic star-planet interaction

2022 **University of Tübingen** Tübingen, GER
Flares on low mass stars: from detection to characterization, and beyond

2022 **American Museum of Natural History** New York City, USA
Flare observations: probing the magnetism of low-mass stars and star-planet systems

2021 **Harvard Smithsonian Center for Astrophysics Exoplanet Pizza Lunch** virtual
Magnetic worlds: What flares tell us about small stars and their planets

Contributed talks (selection)

2026 **CoolStars 23** Tokyo, Japan
Self-roast: Close-in planet induces flares on its host star

2026 **Exoplanets 6** Porto, Portugal
Self-roast: Close-in planet induces flares on its host star

2024 **CoolStars 22** San Diego, CA, USA
Hot Jupiters and stellar eruptions: Planet-induced flares in young exoplanetary systems

2024 **BCool** ASTRON, NL
A fully convective rapid rotator with a high latitude flare and a dim corona

2024 **ESA's Holland-Area Exoplanet Science Meeting** Nordwijk, NL
Hot Jupiters and stellar eruptions: planet-induced flares in young exoplanetary systems

2023 **European Astronomical Society Meeting** Krakow, Poland
Planet-induced flares in Kepler and TESS observations

2023 **The X-ray Universe** Athens, Greece
High-latitude flares on rapidly rotating M dwarfs - a special corona?

2019 **Planet-Star Connections in the Era of TESS and Gaia** Santa Barbara, CA, USA
Stellar magnetic evolution: flaring activity in K2 open clusters

2018 **Dwarf Stars and Clusters with K2** Boston, MA, USA
In search of the flaring-age relation.

Academic service, outreach and civic involvement (selection)

2025→2026 **Postdoc representative @ ASTRON**
Organizing social events and training opportunities for ASTRON early career researchers.

2025 **Podcast**
`50-min podcast` on exoplanets and the search for life for the German Bild der Wissenschaft magazine.

2025→2026 **Lorentz Center Workshop on star-planet interactions**
Initiator and **chair of the SOC** for a 5-day workshop on star-planet interactions at the Lorentz Center in Leiden, NL.

2025 **LOC @ Beyond Earth (ASTRON)**
Conference with 70 participants. Main tasks: Speaker liaison.

2021 →ongoing **Collaboration with various popular science magazines and blogs**
PM Magazin, Newsweek, New Scientist, Leibniz Magazin, Ulrike Boehm's Women in Research blog

2021 →ongoing **Referee @ various astronomical journals**
Nature Astronomy, Astronomy & Astrophysics, AAS Journals, Astronomy and Computing, Journal for Open Source Software, Astronomische Nachrichten

2024 **Weekend workshop @ Sternwarte Berlin**
Planned a weekend long workshop on exoplanets for a small group of interested undergraduate students of all fields.

2023 **Lecturer @ Vienna Summer School of the German Academic Scholarship Foundation**
Week-long interdisciplinary summer school for 20 undergraduate scholars focused on exoplanets, and the relations between exoplanet research, ethics and culture.

2021 →2022 **Student representative @ Internal Scientific Committee at AIP**
Social events, well-being of students at the institute, on-boarding of new students.

2021 **Public talks @ YouTube channel "Urknall, Weltall und das Leben"**
`30-min YouTube video` and `follow-up 12-min YouTube video` (in German), ≈ 40k views total.

Training and internships

2026 **Supervising Junior Researchers Training** Max Planck Society
2-day online training in effective supervision of junior researchers and their projects.

2021-2022 **Career Mentoring** Cusanuswerk
Training in non-violent communication, conflict resolution, diversity awareness, and adaptive leadership.

2019 **3rd Advanced School on Exoplanetary Demographics** Vietri Sul Mare, Italy
Lectures on exoplanet formation, detection, demographics, star-planet interactions.

2018 **Visiting Scientist @ NASA Ames Research Center** Mountain View, CA, USA
Awarded funding for an internship at the Kepler mission headquarters.

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languages **Russian, German (bilingual), English (C2), Dutch (B1), French (B1)**

skills **Python, Fortran, Unix, LaTeX, Git(Hub), CASA, XMM-Newton SAS, XSPEC**

contact **ilin@mps.mpg.de, github/ekaterinailin, homepage**